AI Observability with Grafana Cloud and Claude Code

Why Tracing Matters for AI Agents

- Beyond Text Generation: Tools like Claude Code perform active operations such as file I/O, bash commands and API searches.
- Hidden Complexity: A single user prompt can trigger dozens of underlying tool calls invisible to the end user.
- Full Visibility: Tracing reveals exactly what the agent did, step-by-step latency, and specific success/failure points.
- Industry Standards: Uses OpenTelemetry (OTel), the same open standard used for modern microservices.
- Real-Time Insight: Today, we hook Claude Code's tool calls directly into Grafana Cloud for live monitoring.

Dashboards › Claude Code Tracing Dashboard

Search... ctrl+k

Invite

Tempo Data Source: grafanacloud-scdemo-traces

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Claude Code Tracing Dashboard

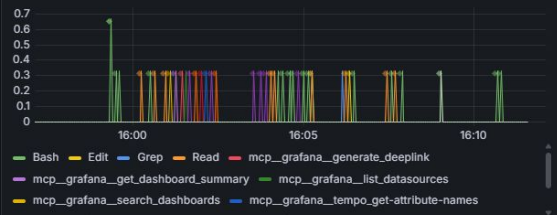
Real-time observability for AI-powered development workflows

OpenTelemetry
Distributed Tracing

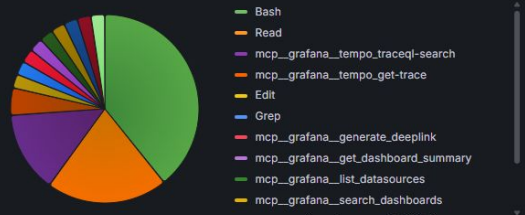
Total Calls	Avg Duration	P95 Duration	Errors	Est. Input Tokens	Est. Output Tokens
43	0.897 ns	0 ns	2	2.09 K	16.9 K

Tool Usage Overview

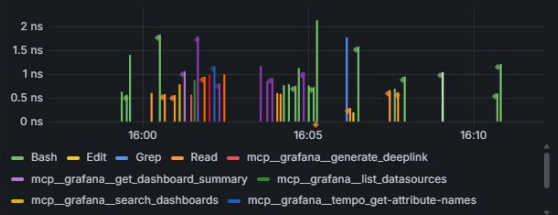
Tool Calls Rate



Tool Usage Distribution

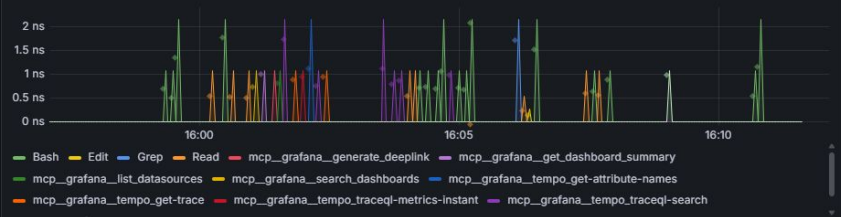
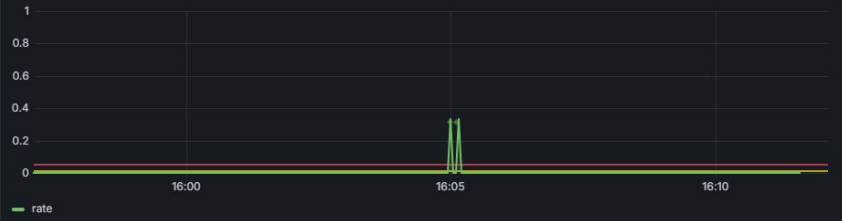


Average Tool Duration



Performance & Errors

P95 Tool Latency



Error Rate by Tool